

The double-stranded lattice sum by frontier elimination

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Abstract

Bio 8 wrote the number of double-stranded reconstructions of a circular sequence as a sum over the lattice points of a box, $\#DS(S, k) = \sum_f N_{\text{seq}}((m + f)/2)$ with f running over $\Lambda^- \cap \{|f| \leq m, f \equiv m \pmod{2}\}$, and enumerated that box by backtracking. The enumeration works at rank $\Lambda^- = 4$ and 12 and dies at 48: $\phi X174$ at $k = 10$ is the blank row of that note's Table 2. The obstruction is not the rank. The circulation condition is one linear equation per vertex, and each equation involves only the four endpoints of one ρ -orbit, so the orbits can be processed in an order while carrying, as state, the partial imbalance at the vertices still open — a product of sparse transfer operators whose cost is governed by the width of the *frontier*, not by the dimension of the lattice. At $k = 10$, where the box lies in a 48-dimensional lattice and has more than two billion points, the elimination needs 50 758 states and 8.4 seconds: $\phi X174$ has exactly 2 481 687 900 box points at $k = 10$. Carrying in addition the partition of the frontier into the classes already joined counts the points with connected support, of which there are 909 103 210; both columns Bio 8 left empty are now filled, and only $\#DS$ itself is not. Two further things follow. Connectivity of $\text{supp } c$, carried in Bio 8 as a separate filter, is not a side condition at all: a balanced digraph has strongly connected weak components, so its out-Laplacian has nullity equal to the number of them, and $t(G_c)$ — hence $N_{\text{seq}}(c)$ — vanishes on a disconnected support of its own accord. And the order that works is not the genome's own: along the reference coordinate the frontier at $k = 10$ is 62 wide and the elimination does not finish, while a greedy order is 26 wide, which is the interleaving of inverted repeats of Bio 1 seen as a width.

Keywords. de Bruijn graph, reverse-complement double cover, circulation lattice, lattice point counting, frontier elimination, transfer matrix, pathwidth, matrix-tree theorem, genome assembly.

1 What is being counted

We keep the notation of [7] exactly. S is a circular sequence of length N , $\bar{S}$ its reverse complement, and $D_k = D_k(S)$ the labelled reverse-complement double cover: its vertices are the k -mers occurring at least twice in the two-strand pool, its labelled arcs the maximal stretches between consecutive such k -mers along each strand, and two arcs carry the same *content* when they spell the same word. Write A for the set of contents, $m(a)$ for the number of labelled arcs of content a , and ρ for the reverse-complement involution, $\rho a = \bar{a}$; then $m(\rho a) = m(a)$, and D_k is balanced.

A circular sequence T is a *double-stranded reconstruction* of S at word length k when $\text{spec}(T) + \text{spec}(\bar{T}) = \text{spec}(S) + \text{spec}(\bar{S})$. With $c = \text{spec}(T) \in \mathbb{Z}_{\geq 0}^A$ this says $c + \rho c = m$ and c is a

circulation; with $f = 2c - m$ it says f lies in the (-1) -eigenlattice $\Lambda^- = \{f \in Z_1(D_k) : \rho f = -f\}$, in the box $|f| \leq m$, in the parity class of m . Hence [7, Thm. 7]

$$\#DS(S, k) = \sum_f N_{\text{seq}}\left(\frac{m+f}{2}\right), \quad N_{\text{seq}}(c) = \frac{t(G_c) \text{Loc}(G_c)}{\prod_a c(a)!}, \quad (1)$$

the sum over $f \in \Lambda^-$ in the box, where G_c is the multidigraph with $c(a)$ copies of each content, $t(G_c)$ its number of spanning arborescences to a fixed root, and $\text{Loc}(G_c) = \prod_v (d_c^+(v) - 1)!$. The division by $\prod_a c(a)!$ passes from labelled arcs to sequences.

Bio 8 enumerated the box by backtracking over a basis of Λ^- with capacity pruning, and recorded the cost: 10 points at $k = 12$ where $\text{rank } \Lambda^- = 4$, 528 at $k = 11$ where $\text{rank } \Lambda^- = 12$, and nothing at $k = 10$, where $\text{rank } \Lambda^- = 48$ and the search does not return.

2 The constraint system and its frontier

The point of this note is that (1) is indexed by the solutions of a very sparse linear system, and that sparsity, not the rank, is what governs the count.

Definition 1 (the orbit variables). ρ acts on A without fixed points except for the self-reverse-complementary contents, which carry $f(a) = 0$ because $\rho f = -f$. Choose one representative a from each free orbit $\{a, \rho a\}$ and let $g_j = f(a_j)$, so $f(\rho a_j) = -g_j$. The box condition is $g_j \in \{-m_j, -m_j + 2, \dots, m_j\}$ with $m_j = m(a_j)$, and f is a circulation exactly when, for every vertex v ,

$$\sum_j B(v, j) g_j = 0, \quad B(v, j) = [v = t(a_j)] - [v = h(a_j)] - [v = t(\rho a_j)] + [v = h(\rho a_j)]. \quad (2)$$

The essential feature is that column j of B is supported on at most four vertices — the two endpoints of a_j and the two of ρa_j — whatever the rank of B may be.

Definition 2 (frontier). Fix an order $\pi = (j_1, \dots, j_R)$ of the orbits. A vertex v is *open after step i* if some j_ℓ with $\ell \leq i$ and some j_ℓ with $\ell > i$ both have $B(v, j_\ell) \neq 0$. Write $\mathcal{F}_i(\pi)$ for the set of open vertices and $w(\pi) = \max_i |\mathcal{F}_i(\pi)|$ for the *width* of π . The minimum of $w(\pi)$ over all orders is the pathwidth of the hypergraph whose vertices are the vertices of D_k and whose hyperedges are the supports of the columns of B .

Theorem 3 (frontier elimination). *Let $\kappa_i(v) = \sum_{\ell > i} |B(v, j_\ell)| m_{j_\ell}$ be the capacity still available at v after step i , and let $X_i = \prod_{v \in \mathcal{F}_i(\pi)} \{-\kappa_i(v), \dots, \kappa_i(v)\}$. Define $T_i: \mathbb{Z}^{X_{i-1}} \rightarrow \mathbb{Z}^{X_i}$ by*

$$(T_i \phi)(y) = \sum_{\substack{g \in \{-m_{j_i}, \dots, m_{j_i}\} \\ g \equiv m_{j_i}(2)}} \sum_{\substack{x \in X_{i-1} \\ x + gB(\cdot, j_i) \text{ restricts to } y \\ \text{and vanishes at every vertex closing at } i}} \phi(x).$$

Then

$$|\Lambda^- \cap \{|f| \leq m, f \equiv m(2)\}| = (T_R \cdots T_1 \delta_0)(\cdot),$$

and the whole product costs $O(\sum_i |X_{i-1}| (m_{j_i} + 1) |\mathcal{F}_i|)$ integer operations, with

$$|X_i| \leq \prod_{v \in \mathcal{F}_i} (2\kappa_i(v) + 1) \leq (2 \max_v \kappa_i(v) + 1)^{w(\pi)}.$$

Proof. A point of the box is an assignment $g \in \prod_j \{-m_j, \dots, m_j\}$ of the right parity satisfying (2). Process the orbits in the order π and record, after step i , the partial sums $x(v) = \sum_{\ell \leq i} B(v, j_\ell) g_{j_\ell}$ at the open vertices. A vertex v that is not open after step i receives no further contribution, so (2) at v is decided then and there: it is the condition, imposed in T_i , that $x(v) = 0$ at every closing vertex. A vertex not yet touched has $x(v) = 0$ and need not be recorded. Two assignments of $g_{j_1}, \dots, g_{j_i}$ with the same restriction to $\mathcal{F}_i$ therefore admit exactly the same completions, which is the Markov property that makes the transfer operators well defined and their product the count. The capacity bound holds because $|x(v)| = |\sum_{\ell > i} B(v, j_\ell) g_{j_\ell}| \leq \kappa_i(v)$ for any state that has a completion, so states outside X_i may be dropped. $\square$

Remark 4 (what the cost does and does not depend on). Theorem 3 contains no reference to rank Λ^- . The backtracking of [7] has cost proportional to the number of *nodes* of a search tree of depth R , which for a lattice of rank r is at least the number of points and so at least exponential in r ; the elimination has cost proportional to the number of *states*, and a state is a vector of imbalances on the frontier. At $k = 10$ these are 2 481 687 900 and 50 758.

Proposition 5 (product-form weights). *Let u_j be functions on the values of g_j and h_v functions on $\mathbb{Z}_{\geq 0}$, and*

$$w(c) = \prod_j u_j(g_j) \prod_v h_v(d_c^+(v)).$$

Then $\sum_f w(c_f)$ over the box is computed by Theorem 3 after enlarging the state at each open vertex by its accumulated out-degree. In particular $\text{Loc}(G_c)$ and $\prod_a c(a)!$ are of this form.

Proof. $d_c^+(v) = \sum_{a: t(a)=v} c(a)$ accumulates one orbit at a time exactly as $x(v)$ does, and is final when v closes, where h_v may be applied; u_j multiplies into the transfer operator at step j . Loc takes $h_v(d) = (d-1)!$ and $u_j = 1$, while $\prod_a c(a)!$ takes $h_v = 1$ and $u_j(g) = (\frac{m_j+g}{2})!(\frac{m_j-g}{2})!$. $\square$

3 Connectivity is the matrix-tree determinant

Bio 8 carried, beside the box conditions, a predicate: the support of c must be connected, or the point contributes nothing. The predicate is real but it is not an extra condition.

Lemma 6 (connectivity is not a side condition). *Let $c \in \mathbb{Z}_{\geq 0}^A$ be balanced, $V_c = \text{supp } c$ the set of vertices it meets, L_c the out-degree Laplacian of G_c on V_c and $\tilde{L}_c$ the matrix L_c with one row and the corresponding column deleted. Then $\det \tilde{L}_c = t(G_c)$, and*

$$\det \tilde{L}_c = 0 \iff G_c \text{ is disconnected.}$$

Consequently $N_{\text{seq}}(c) = 0$ whenever $\text{supp } c$ is disconnected, and the connectivity predicate may be deleted from (1) without changing the sum.

Proof. $\det \tilde{L}_c = t(G_c)$ is the matrix-tree theorem for digraphs [11]. A balanced digraph has $d^+ = d^-$ at every vertex, so each of its weakly connected components is Eulerian and hence strongly connected; therefore the kernel of L_c is spanned by the indicators of the components and $\dim \ker L_c = r$, the number of them. Deleting one row and one column drops the nullity by at most one, so $r \geq 2$ forces $\det \tilde{L}_c = 0$, while $r = 1$ gives $t(G_c) \geq 1$. $\square$

Remark 7. The gain is conceptual rather than computational: as a filter, connectivity is a union-find pass and the determinant is a cubic elimination, so one still tests connectivity first. What Lemma 6 removes is the impression that (1) is a lattice sum *subject to a combinatorial constraint*. It is a lattice sum, full stop; the constraint is already inside N_{seq} . This is also the answer, in the only form it has, to the question of [7] for an algebraic criterion of connected support avoiding graph traversal: the criterion is the vanishing of a determinant whose entries are linear in c .

Connectivity can also be carried *inside* the elimination, and then it counts.

Theorem 8 (connected points). *Enlarge the state of Theorem 3 by (i) the partition of $\mathcal{F}_i$ into the classes connected by arcs already chosen, (ii) for each class, whether it meets a chosen arc, and (iii) the number of classes that have left the frontier while carrying a chosen arc. Prune whenever (iii) exceeds one, and accept only states with (iii) equal to one and empty frontier. The resulting count is the number of box points with connected support. The state count is at most $|X_i| \cdot B(|\mathcal{F}_i|) \cdot 2^{|\mathcal{F}_i|} \cdot 2$ with B the Bell numbers.*

Proof. A class leaves the frontier only when no vertex of it is open, and after that no arc can join it to anything; so a class that leaves while meeting a chosen arc is a finished connected component of $\text{supp } c$, and conversely every component is finished exactly once, at the step where its last vertex closes. A support is connected iff it has exactly one component, which is condition (iii); classes meeting no chosen arc are not vertices of $\text{supp } c$ and must be ignored, which is what (ii) records. Since the class of a vertex and its flag are determined by the chosen arcs among the processed orbits, the state is again Markov. $\square$

Remark 9 (the connectivity state is cheaper than it looks). The Bell factor in Theorem 8 is a gross overestimate, because the rule “at most one finished component” forbids most partitions from ever arising: at $k = 12$ and $k = 11$ the connected elimination peaks at 8 and 136 states against 4 and 20 for the plain one. It is nevertheless the factor that decides the cost at $k = 10$, where a frontier of 26 vertices is small for imbalances and large for partitions: 9 749 212 states against 50 758, and seventy-eight minutes against eight seconds. $B(26)$ is 5×10^{19} , so what is actually reached is a vanishing fraction of the bound.

4 The genome’s own coordinate is not the good order

Theorem 3 turns the problem into the choice of π . There is an obvious candidate, and it is the one the biology suggests: take the orbits in the order of their position along the reference genome, so that the elimination sweeps the chromosome once. It is bounded, and it is not good.

Proposition 10 (the coordinate order). *Order the orbits by the position of their first occurrence on S . Then $\mathcal{F}_i$ is contained in the set of repeated k -mers having occurrences in orbits on both sides of step i ; in particular, if no more than ω repeat classes straddle any coordinate, the elimination runs in $O(R(2 \max \kappa + 1)^\omega)$ and $\#DS$ ’s box is counted in time polynomial in N for fixed ω .*

Proof. Immediate from Definition 2: the vertices met by orbit j are endpoints of a_j and ρa_j , and a vertex is open only between the first and last orbit meeting it. $\square$

The hypothesis is the point. Table 1 gives the widths.

k	$ V $	contents	orbits	rank Λ^-	coordinate width	greedy width
12	7	14	7	4	6	6
11	30	53	26	12	18	10
10	125	222	111	48	62	26

Table 1: The width of the two orders. Along the genome the frontier at $k = 10$ is 62 vertices wide and the elimination does not finish — it passes 1.9 million states and is abandoned; the greedy order is 26 wide and finishes in 8.4 seconds with 50 758 states. Check (O).

k	rank Λ^-	box		connected		% conn.	#DS
		points	states	points	states		
13	0	1	—	1	—	100	1
12	4	10	4	8	8	80.0	10
11	12	528	20	218	136	41.3	864
10	48	2 481 687 900	50 758	909 103 210	9 749 212	36.6	—

Table 2: The blank row of [7, Table 2], filled in both of its counting columns. The rows for $k = 12$ and $k = 11$ reproduce that table exactly, by the elimination and independently by the backtracking enumerator; the box count at $k = 10$ is confirmed by pinning one coordinate and summing the elimination over its values. “states” is the peak number of frontier states: 8.4 seconds for the box, 78 minutes and 10 GB for the connected count. Checks (T), (K), (W), (M).

Remark 11 (why the coordinate order fails). A vertex stays open from its first to its last incident orbit, so the coordinate width at a position x counts the repeat classes straddling x . Two things inflate it. First, ρ pairs an arc of S with an arc of $\bar{S}$, and the two lie at reflected coordinates, so an orbit is not local on the reference at all: every orbit near the origin of S is also an orbit near the end of $\bar{S}$. Second, and this is the substance, inverted repeats of a real sequence *interleave* — that is the content of the interleaving theorem of [4], where the crossing pattern of repeats is exactly what the sandpile group of the de Bruijn graph records. Interleaved repeats are nested intervals crossing each other, and a family of pairwise crossing intervals forces every one of them to be open at the crossing point. The coordinate width is therefore a measure of interleaving, and a genome with interleaved inverted repeats has no narrow sweep. The greedy order abandons the chromosome and follows the constraint hypergraph instead.

5 $\phi\mathbf{X174}$ at $k = 10$

$\phi\mathbf{X174}$ at $k = 10$ has 125 vertices, 222 contents, 111 free ρ -orbits and rank $\Lambda^- = 48$; the box $\Lambda^- \cap \{|f| \leq m, f \equiv m \pmod{2}\}$ contains exactly

$$2\,481\,687\,900$$

points. Three remarks on the number. It is exact and integral throughout — the elimination adds integers and never divides. It is confirmed independently: pinning one orbit variable to each of its admissible values in turn and summing the resulting eliminations, for two different orbits, returns the same total, and the two orbits have disjoint schedules. And it is not reachable by enumeration: 2.48×10^9 points at the rate at which [7] enumerates 528 of them would be

several months, and the relevant comparison is that the whole count took 8.4 seconds because it never listed a point.

Remark 12 (the connected fraction). The proportion of box points with connected support falls with k : 100%, 80%, 41.3%, 36.6% at $k = 13, 12, 11, 10$. This is what one expects and it is worth saying why. A point chooses one arc from each free ρ -orbit, so the support of c has close to $|A|/2$ arcs on $|V|$ vertices whatever the point; at $k = 10$ that is about 111 arcs on 125 vertices, which is sparse enough that a random choice usually falls apart. What the elimination shows is that 36.6% of them do not, and that the surviving set is still of size 9×10^8 — so the failure to reach #DS at $k = 10$ is not for want of a small enough index set.

Remark 13 (what is still missing, exactly). Of the three counting columns of Table 2 only #DS is now empty at $k = 10$, and it is empty for a reason of a different kind from the other two. Those were questions of state size: Theorem 3 settles the first in seconds and Theorem 8 the second in an hour and a quarter.

The #DS column is a question of principle. By Proposition 5 the elimination sums any weight that factors into per-orbit and per-vertex terms, and two of the three factors of N_{seq} do: $\text{Loc}(G_c)$ is a product over vertices of a function of the out-degree, and $\prod_a c(a)!$ a product over orbits. The third, $t(G_c)$, does not factor over any bounded frontier — it is a determinant of order up to $|V_c|$, a global function of the whole support — and it is the one that carries the count. Evaluating it point by point is cheap per point and hopeless in aggregate: 9.09×10^8 determinants of order up to 125. So #DS($\phi X174, 10$) remains out of reach, and it is now clear that what stands in the way is the arborescence factor and not the lattice — the index set is known exactly, and it is the summand that cannot be carried. Bio 8’s Table 2 gives the two ends of the interval it lies in: the single-strand count $N_{\text{seq}}(S) + N_{\text{seq}}(\bar{S}) = 8\,610\,708\,632$ is a lower bound, since $\text{spec}(S)$ and $\text{spec}(\bar{S})$ are two of the connected points, and $\text{ec}(D_{10}) = 1.368 \times 10^{36}$ is an upper one.

6 Verification

`ds_transfer_matrix.py` is standalone — Python and the standard library, no `numpy`, no `sympy` and no other file of the series — and rebuilds D_k from the definition rather than importing it, so that check (D) is a genuine cross-check of Bio 8’s Table 1 and not a tautology. Exact integer arithmetic throughout; the determinants of Lemma 6 are computed over $\mathbb{Q}$ with `fractions`.

7 Scope, provenance, and what is open

Scope. Theorem 3, Proposition 5, Lemma 6 and Theorem 8 hold for any circular S and any k . Proposition 10 is a bound, and Table 1 shows it is not a good one on real sequence. Section 5 is a computation on one named genome and claims nothing beyond it. Nothing here improves (1) itself, which is Bio 8’s.

Provenance. The lattice sum, Λ^- , its rank by the Hopf trace formula, and the $k = 12$ and $k = 11$ values are [7]; the double cover and its balance are [5], its deck involution as an anti-automorphism [6], the interleaving of repeats [4]. The BEST theorem is [1, 10] and the matrix-tree theorem for digraphs is [11]. Frontier elimination is not new: it is the standard dynamic-programming-over-a-path-decomposition, and in the form used in Theorem 8 — carrying a partition of the frontier so as to count connected subgraphs — it is the frontier-based search

key	claim
(D)	the double cover rebuilt from the definition reproduces $ V $, $\#\text{fix}_V$, $ A $, $\#\text{fix}_A$, $\sum m$ and $\text{rank } \Lambda^-$ of [7, Table 1] at $k = 10, 11, 12$
(T)	the frontier elimination and the backtracking enumerator return the same box count, 10 and 528
(C)	on all 538 box points at $k = 12, 11$: $\text{supp } c \text{ connected} \iff \det \tilde{L}_c \neq 0$ — Lemma 6
(W)	summing N_{seq} over the box returns 8 and 218 connected points and $\#\text{DS} = 10$ and 864
(O)	coordinate width 6, 18, 62 against greedy width 6, 10, 26; at $k = 10$ the coordinate order does not finish
(K)	$\phi X174$ at $k = 10$: 2 481 687 900 box points, and the same total after pinning each of two orbits
(N)	the connectivity-class elimination returns Bio 8’s 8 and 218
(M)	$\phi X174$ at $k = 10$: 909 103 210 box points with connected support

Table 3: The eight checks. All pass. Check (M) takes about eighty minutes and 10 GB; the flag `--quick` skips it and the rest of the battery runs in six minutes.

of [8], itself descended from Knuth’s `SIMPAT`H [9]; the width parameter is pathwidth [3]. What is specific here is the observation that the double-stranded reconstruction count is indexed by a system whose columns have support four, so that the frontier is narrow for reasons having nothing to do with $\text{rank } \Lambda^-$, and the identification of the coordinate order as the wrong one. For counting lattice points in a polytope of *fixed* dimension there is Barvinok’s algorithm [2]; it does not apply here, the dimension being the thing that grows.

Open. *First*, $\#\text{DS}(\phi X174, 10)$, for which something must be done about $t(G_c)$; the natural attempt is to carry a partial Schur complement of L_c on the frontier, which is a matrix state and not a scalar one, and whether that can be made to fit is the real question this note leaves. An order chosen to keep partitions few rather than imbalances small would help both it and Theorem 8, whose cost at $k = 10$ is entirely the partition factor. *Second*, the width of a bacterial genome. The constraint hypergraph of *E. coli* at $k = 31$ falls apart into many small blocks, one per repeat cluster, and the elimination factors over them; whether the largest block stays narrow at the word lengths where the double cover is interesting is not known to us. *Third*, whether the coordinate width of Proposition 10 is, for a random sequence with a prescribed inverted-repeat structure, a computable function of that structure — it would make “how hard is double-stranded assembly here” a statistic of the genome.

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